

Internship Project Guidelines

VeCAD Elite 2026 | Embedded System | ST Edge AI Group

AI, Edge AI and AIoT Applications for Smart Building Energy Systems

General Project Theme

The three internship projects focus on the application of **Artificial Intelligence (AI)**, **Edge AI** and **AIoT** to smart building and classroom energy systems.

To ensure that the projects are safe and accessible within a short internship period, a **Low-Voltage Mini Classroom (LV-MC)** concept is used where applicable.

Important Safety Requirement

You must **not connect your project directly to any building electrical supply, 240 V AC mains, wall socket, distribution board, classroom lighting circuit, air-conditioning system, or other building electrical infrastructure.**

All energy-related experiments must use **safe, low-voltage DC loads**, such as a 5 V LED lamp and a small 5 V DC fan.

PROJECT 1

TinyML-Based Classroom Occupancy Estimation for Energy-Efficient Building Applications

Development Board: NUCLEO-L476RG

Project Area: TinyML / Edge AI / Smart Building

Recommended Duration: 1 month

Effective Technical Work: Approximately 10 working days

1. Project Overview

Energy consumption in classrooms is influenced by the number of occupants and the usage of electrical equipment. Occupancy information can therefore be useful in identifying opportunities for improving energy efficiency.

In this project, you will develop a **TinyML-based classroom occupancy estimation system** using low-cost sensors and the **STM32L476RG** microcontroller.

You will collect sensor data representing different classroom occupancy conditions, develop a lightweight machine-learning model, and deploy the trained model to the STM32 microcontroller.

The completed system should be capable of estimating the occupancy condition of a classroom in real time and providing a simple indication of its potential energy-saving status.

The project is designed as a **prototype for smart-building applications**. You are not required to access or modify any actual building energy-management system.

2. Project Objectives

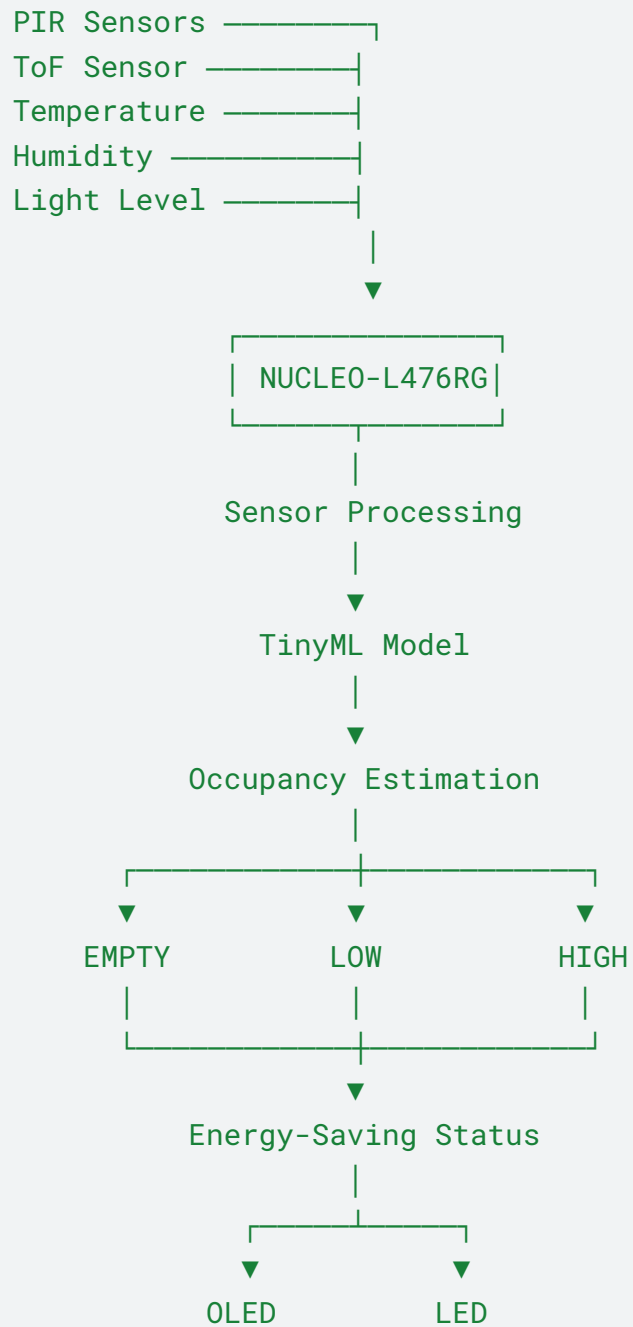
By completing this project, you should be able to:

1. Interface multiple sensors with an STM32 microcontroller.
 2. Collect and organize sensor data for machine-learning purposes.
 3. Label and prepare a dataset for occupancy classification.
 4. Develop a lightweight machine-learning classification model.
 5. Evaluate the performance of the trained model.
 6. Convert and deploy the model using **STM32Cube.AI**.
 7. Implement real-time TinyML inference on the STM32L476RG.
 8. Display the estimated occupancy condition using an OLED display and/or LED indicators.
 9. Explain how occupancy information can support energy-efficient building applications.
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3. Proposed System

The overall system should follow the architecture below:

None



4. Suggested Occupancy Classes

Three occupancy classes should be sufficient for the initial implementation.

Class	Description
Empty	No person is present or detected
Low	A small number of people are present
High	Several people are present

The objective is **not necessarily to count every individual accurately**. The emphasis should be on determining an occupancy condition that can subsequently be used for energy-efficiency decisions.

5. Required Hardware

Already Available

Component	Quantity
NUCLEO-L476RG	1

Additional Components

Component	Quantity
PIR motion sensor	2
VL53L0X ToF sensor	1
BME280 temperature/humidity sensor	1
BH1750 ambient-light sensor	1
SSD1306 OLED display	1
LEDs	1 set
Push buttons	2
Buzzer	1
Breadboard	1
Jumper wires	1 set
USB cable	1

6. System Development

You should complete the project through the following stages:

Stage 1 — Hardware Setup

- Configure the NUCLEO-L476RG.
- Interface the PIR sensors.
- Interface the ToF sensor.
- Interface the environmental sensor.
- Interface the ambient-light sensor.
- Verify sensor readings individually.

Stage 2 — Data Collection

You should collect data under different occupancy conditions.

For example:

None

Condition 1 → Empty

Condition 2 → Low occupancy

Condition 3 → High occupancy

Each record should contain the relevant sensor measurements and the corresponding occupancy label.

Stage 3 — Machine Learning

You should:

1. Clean the collected data.
2. Analyze the sensor characteristics.
3. Select appropriate input features.
4. Divide the dataset into training and testing data.
5. Train a classification model.
6. Evaluate its performance.

A small neural network may be used as the primary Edge AI model. Other lightweight models may be used for comparison.

Stage 4 — Edge AI Deployment

The trained model should be:

- optimized where appropriate,
- quantized if suitable,
- converted using STM32Cube.AI,
- integrated into the STM32 firmware.

Stage 5 — Real-Time Testing

The final system should be tested using different occupancy conditions.

The OLED/LED output should indicate the estimated occupancy condition.

7. Expected Deliverables

You should submit:

Hardware

- Functional sensor system.
- Working NUCLEO-L476RG prototype.
- OLED/LED occupancy indicator.

Software

- STM32CubeIDE project.
- Sensor-interface firmware.
- Machine-learning training files/notebook.
- Dataset.
- Trained AI model.
- STM32Cube.AI implementation.

Technical Documentation

- System block diagram.
- Circuit schematic.
- Dataset description.
- AI methodology.
- Model-performance results.
- Edge AI implementation.
- Testing results.
- Limitations and possible improvements.

Final Demonstration

You should demonstrate:

Change in occupancy → sensor response → TinyML inference → occupancy classification → energy-saving indication
